Supplementary Material for [Experiment, Analysis, and Benchmark] Recovering Vehicle Trajectories from CAN Signals: A Large-Scale Experimental Study

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***Abstract*—This document accompanies the paper with the material that supports its reproduction: the structure of the released package and the commands that rebuild every table and figure (S1), the benchmark definitions in full (S2), the reproduction details of the six published methods and of the benchmark baselines (S3), the complete result tables behind the paper's findings, including the figures that the page limit left out (S4), the full protocol, prompts and per-configuration results of the supplementary agent study (S5), and further recovered windows with their camera frames (S6). Code: https://github.com/WangYuHang-cmd/GPS--CAN. Data, scores, configurations, logs and weights: https://huggingface.co/datasets/CVPRICCVICML/GPS_Reconstruction.**

I. S1. Reproduction package

The code repository holds the `cantraj` package (data layer, signal whitelist, causal replay, metrics and scoring, baselines, map graphs, learned motion models, the agent study), the scripts that run every experiment and build every table and figure, the frozen configuration (`configs/`: benchmark thresholds and manifest hashes, signal registry with provenance classes, signal groups, protocols, windows, agent prompts and budgets), the specification of benchmark v1 and the test suite (139 tests on synthetic data). The data package holds the dataset (3,940 drives of 454 drivers: re-decoded CAN event streams at logged times, the aligned GNSS reference with the first and last two minutes of every drive cut, quality metadata and driver-disjoint splits), the benchmark manifests (episodes, map-start rule, tile list with md5 and snapshot date, cohorts), every per-episode score of every run, the run registry with code, scorer and tile hashes, the agent study's configuration and conversation logs, the map tiles and the learned models' weights.

a) Rebuilding the tables: From the released per-episode scores alone: copy the package's `tables/` and `reports/` into `reports/icde/`, then run `icde_tables.py --extras a3 pairs topk signal cohort cost robust robust_map`, `icde_area_check.py`, `icde_extra.py` (and `--what supp`), `icde_spy_dewri_diag.py`, `failure_analysis.py`, `icde_figures.py` and `icde_macros.py`. Every number of the paper is a macro in `macros_numbers.tex` written by the last script.

b) Rerunning the methods: `run_baseline.py` with the v1.1 flags (`--version v1.1 --manifest episode_manifest_v1.parquet --eligible-only --tiles-dir maps/tiles_v2 --map-episodes episode_map_v11.parquet`), one chain script per study (`scripts/run_icde_*.sh`), then `score.py --run <id>`. Prediction files are regenerated, not shipped. The agent study needs a served Qwen3-14B (revision pinned in the configuration) behind an OpenAI-compatible endpoint; `scripts/h100_setup.sh` installs and serves it, `run_icde_agent.sh` runs the cache, the selector, the three seeds, the scoring and the tables.

II. S2. Benchmark definitions

a) Episodes and eligibility: Windows of 60, 300 and 900 s and whole drives tile each drive from its first reference fix; a window starts at an anchor fix (speed ≥ 3 m/s, bearing accuracy within 5°, horizontal accuracy under 10 m, CAN speed ≥ 2.5 m/s) and is eligible when it has at least 0.8 reference fixes per second, no reference gap over 30 s and a path of at least 100 m. Drives with an unconfirmed vehicle platform and the controlled supplement stay out of the main board. The map-start rule of v1.1 requires a drivable edge within 400 m of the start in the visible map; windows that fail it are reported separately and never counted against a method.

b) Splits: Driver-disjoint folds: train, validation, test and few-shot held-out (776 held-out drives of 94 drivers). Parameters and models use train and validation only.

c) Signals and provenance: Signal sets S1 (speed), S2 (speed and pedals), S3 (speed and steering) and S4 (all); groups G1 to G8 whitelist columns and static parameters; every signal carries a provenance class (bus-read, derived in the vehicle software, logger sensor, GPS-informed), and class-F references are never in the CAN-only board.

d) Protocols: Online or offline access, map or no map, start heading given or withheld; the visible map is the disc of radius $5\,\mathrm{km} + 40\,\mathrm{m/s} \cdot D$ (at most 60 km) around the start, enforced per window.

e) Metrics: ADE and FDE at the reference seconds, coverage within 10 to 500 m, time-free path recall and precision at 25, 50 and 100 m, route recovery (path recall at 50 m ≥ 0.8),

terminal excursion and tail-lost flags; an expected window without output is a failure (ADE $+\infty$, rates 0); medians carry driver-clustered bootstrap intervals (1,000 resamples); paired differences are computed on common windows.

III. S3. Methods

The six published methods were re-implemented from their papers and, where available, checked against the authors' released code on the authors' own samples (OPD-II: 8 of 59 logs within a median of at most 0.33 m of the unmodified script; MacroTracking: the sample trace within a median of 7.1 m of the committed trajectory). Adaptations are named as such in Table I: Elastic Pathing and Dewri et al. with the start heading (start edges within 45° of it), CAN-Trace and SPy with a 5 km square centred on the start, OPD-II with a strictly online read-out. Parameters left open by the papers were set on training and validation drives only. The benchmark baselines are dead reckoning (nominal, calibrated, with an online offset), a route prior and route retrieval, a map particle filter (2,000 particles, 10 Hz ticks, heading and turn noise of 15 and 20°) and the offline map beam search (beam 200, ten candidates, heading and turn widths 15 and 20°, U-turn weight 0.05, re-anchoring gain 0.3, length slack 2 % with a 10 m floor, dead-end weight 0.1, 50 m start radius, 10 m merge bins, temperature 1), as given in Algorithm 1 of the paper. The learned motion models (a GRU and a physics residual, three seeds) are trained on the training fold only with validation early stopping.

IV. S4. Complete results

V. S5. Agent study

The supplementary agent study (Section VI of the paper) compares, on one five-minute window per held-out drive (268 test windows of 42 drivers, 391 few-shot windows of 50 drivers), the offline beam (A), a supervised selector over the three reconstructors (B) and a Qwen3-14B tool-calling controller (C). The protocol was frozen after a validation pilot (`reports/agent_protocol_v1.md`): model revision, sampling, prompt, tool schema, budget, selector weights and cohort hashes are recorded there; every window of every seed received a valid choice. Tables XXII to XXV give every row.

a) Cost: The reconstructed serial cost of a window is the model's wall time plus the measured single-worker time of the tools it ran plus the tool execution time. On 10 validation windows every reconstructor call was executed uncached and single-stream: median end-to-end time 27.7 s against a reconstructed serial cost of 23.4 s on the same windows (ratio 1.08, 90th percentile 1.68).

b) System prompt (verbatim):

```
You are the controller of a vehicle-trajectory recovery
system. One five-minute driving window is given. The
system knows only the vehicle's CAN signals (speed and
steering-wheel angle), the start position (the origin),
the start heading, static vehicle parameters and the road
graph within the visible disc around the start. No GPS
positions, no true route and no scores are available to
you or to the tools; your job is to pick the candidate
path most likely to be the vehicle's true path.

Tools: run_reconstructor(name) runs one of three existing
reconstructors (beam, pf, ep) and returns its ranked
candidate paths with diagnostics;
inspect_candidate(candidate_id) gives the full diagnostics
of a candidate you have already seen;
query_local_graph(candidate_id, distance_m) describes the
road junction nearest to that distance along a candidate
you have seen; submit_choice(candidate_id, reason) ends
the task. Every reconstructor can be run at most once. You
must finish with submit_choice for a candidate id that a
reconstructor returned to you; nothing else is accepted.

Diagnostics (all computed from the CAN signals, the given
start heading and the road graph): path_len_m is the
candidate's length; len_ratio_dev is |path length /
CAN-integrated distance - 1|; hd_abs_rms_deg is the RMS
difference between the candidate's road bearing and the
dead-reckoned heading (start heading plus integrated
steering) over the moving seconds; hd_drift_rms_deg is the
same after removing the best constant offset; turns_path
and turns_dr count turns (bearing change above 45 degrees
within 40 m) along the candidate and in the dead-reckoned
heading, turns_unmatched counts turns present in one but
not the other (same sign, within 50 m of path distance);
uturns counts reversals; overlap_with_called_top1 is the
share of the candidate within 15 m of the Top-1 of another
reconstructor you have run (unknown for reconstructors you
have not run); end_dist_to_called_top1_m compares end
points; n_called_tools_agree counts other run
reconstructors whose Top-1 overlaps this candidate by at
least 0.8. Positions are metres east and north of the
start; compass headings are degrees clockwise from north.
Dead reckoning accumulates error, so small heading or turn
disagreement does not by itself rule out a candidate;
large unmatched turns or a path much shorter or longer
than the CAN distance are strong evidence against it.

Budget: at most 4 decision turns, at most 7 tool calls
before the final submit_choice, and a limited number of
generated tokens. You may request several tool calls in
one turn (for example two reconstructors at once); each
call counts toward the 7. Be concise. When the budget
forces it, you will be asked to submit immediately.
```

c) Tool schema and budget:

```
run_reconstructor(name)
Run one trajectory reconstructor on this window (each
reconstructor can be run once; a repeat returns the same
result and still counts). Returns its ranked candidate
paths with observable diagnostics.
inspect_candidate(candidate_id)
Detailed diagnostics of one candidate you have already
seen: heading consistency, turn matching against the
dead-reckoned heading, end position, overlap with the
Top-1 of the other reconstructors you have run, and where
it diverges from them.
query_local_graph(candidate_id, distance_m)
Road junction nearest to a given distance along a
candidate you have seen: arrival heading, the exits
(relative turn, length, road class, one-way), which exit
the candidate took, and the dead-reckoned heading change
around that moment.
submit_choice(candidate_id, reason)
Final answer: the id of one candidate you have seen. The
program outputs that candidate's full trajectory.
budget: 4 decision turns, 7 tool calls + 1 submit, 4096
completion tokens (per turn 2048, reserve 256), virtual
clock 600 s
```

VI. S6. Recovered windows

References

[1] A. Gazdag, S. Lestyán, M. Remeli, G. Ács, T. Holczer, and G. Biczók, "Privacy pitfalls of releasing in-vehicle network data," *Vehicular Communications*, vol. 39, p. 100565, 2023.

TABLE I

METHODS AND THE INFORMATION THEY USE. p_0: START POSITION, θ_0: START HEADING. ONLINE METHODS ANSWER EACH QUERY FROM THE SIGNALS UP TO THAT TIME. MAP: "OSM DISC" IS THE EPISODE'S VISIBLE DISC (SECTION II). DR: DEAD RECKONING, PF: PARTICLE FILTER, GRU: GATED RECURRENT UNIT, S1 TO S4: SIGNAL SETS (SECTION III).

method	source	signals	priors	map	mode	training data	output
Hold start	baseline	none	p_0	none	online	none	position
Straight line	baseline	speed	p_0, θ_0	none	online	none	position
Steering DR	baseline	speed, steering	p_0, θ_0	none	online	none / calib.[a]	position
Start prior	baseline	speed	p_0, θ_0	none	online	train GPS (statistics)	position
Route retrieval	baseline	speed	p_0, θ_0	none	online	train GPS (routes)	position
Learned GRU	baseline	speed, steering	p_0, θ_0	none / OSM disc	online	train GPS	position
Learned physics residual	baseline	speed, steering	p_0, θ_0	none / OSM disc	online	train GPS	position
Map PF	baseline	speed (+steering)	p_0 (+θ_0)	OSM disc	online	none	position
Map beam	baseline	speed (+steering)	p_0 (+θ_0)	OSM disc	offline	none	top-10 paths
MacroTracking [1]	re-impl.	speed, steering	p_0, θ_0	OSM disc	online	none / calib.[a]	position
OPD-II [2]	re-impl.	speed, steering	p_0, θ_0	OSM, Meili[e]	delayed $\leq$3 s[b]	none / calib.[a]	position
Elastic Pathing [3]	re-impl.	speed (1 Hz)	p_0[c] (+θ_0)	OSM disc	offline	val (2 values)	top-n paths
CAN-Trace [4]	re-impl.	speed, accel. pedal	area[c] / p_0	OSM disc	offline	none	top-10 paths
Dewri et al. [5]	re-impl.	speed (1 Hz)	p_0[c] (+θ_0)	OSM disc	offline	none	top-10 paths
SPy [6]	re-impl.	steering	area[c] / p_0 (+θ_0)	OSM disc	offline	val (2 values)	path
Signal-set GRU (S1 to S4)	study	per set	p_0, θ_0	none / OSM disc	online	train GPS	position
Curvature DR (class F)	reference	speed, localizer curv.	p_0, θ_0	none / OSM disc	online	GPS-informed[d]	position

[a] variant with training-fold steering calibration. [b] each position is fixed when its 3 s batch is map-matched; a strictly online read-out is also reported. [c] native prior first, then our adaptations to the shared protocols; "area" is a 5 km square centred on the known start. [d] the localizer used GPS in the software versions of this data; reference only. [e] Meili matches within its Valhalla region tiles, 200 m around the dead-reckoned trace, and is not cut to the episode's disc.

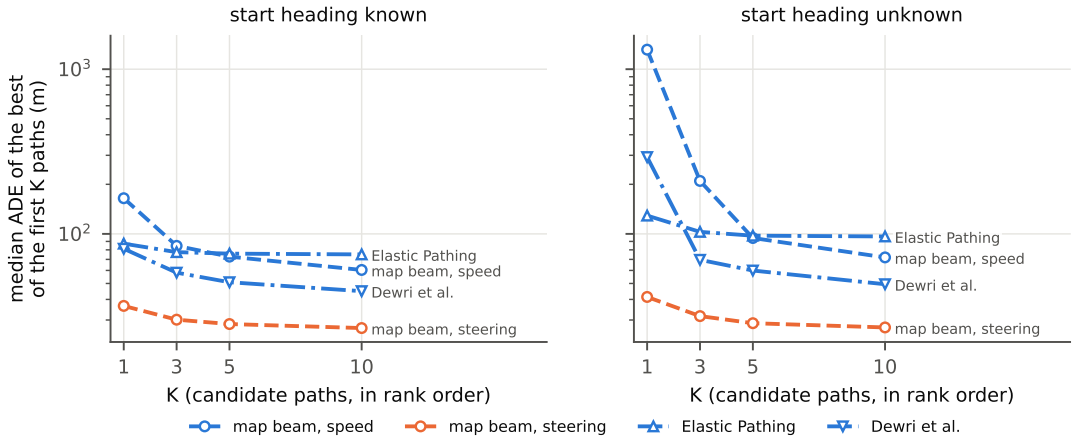


Fig. 1. Best of the first K candidates (whole-path oracle median ADE and per-second union coverage within 100 m) for the offline beam with and without steering and for Elastic Pathing and Dewri et al., five-minute windows.

[2] T. Bianchi, A. Brighente, M. Conti, and A. Valori, "Your car tells me where you drove: A novel path inference attack via CAN bus and OBD-II data," in *Proceedings of the 10th IEEE European Symposium on Security and Privacy (EuroS&P)*, 2025, pp. 113–132.

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[4] X. Lin, B. Ma, X. Wang, G. Yu, Y. He, W. Ni, and R. P. Liu, "CAN-Trace attack: Exploit CAN messages to uncover driving trajectories," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 3, pp. 3223–3236, 2025.

[5] R. Dewri, P. Annadata, W. Eltarjaman, and R. Thurimella, "Inferring trip destinations from driving habits data," in *Proceedings of the 12th ACM Workshop on Privacy in the Electronic Society (WPES)*, 2013, pp. 267–272.

[6] M. D. Pesé, X. Pu, and K. G. Shin, "SPy: Car steering reveals your trip route!" *Proceedings on Privacy Enhancing Technologies*, vol. 2020, no. 2, pp. 155–174, 2020.

TABLE II

COMPLETE BOARD, 5 MIN WINDOWS, BENCHMARK V1.1: EXPECTED WINDOWS n, FAILURE SHARE (NO OUTPUT), FAILURE-AWARE MEDIAN ADE IN METRES WITH THE 95 % DRIVER-CLUSTERED CI, AND THE ROUTE RECOVERY RATE (PATH RECALL AT 50 M $\geq$ 0.8), TEST AND FEW-SHOT DRIVERS.

method	protocol	n	fail %	median ADE m [CI]	route %	n	fail %	median ADE m [CI]	route %
hold start	O-N-Hp	1,917	0.0	3,314 [2,113, 4,313]	0	1,963	0.0	2,709 [2,232, 3,562]	0
straight line (speed)	O-N-Hp	1,917	0.0	735 [644, 911]	9	1,963	0.0	825 [781, 951]	7
start prior (speed, train stats)	O-N-Hp	1,917	0.0	1,068 [920, 1,119]	2	1,963	0.0	1,062 [946, 1,162]	2
route retrieval (speed, train routes)	O-N-Hp	1,917	0.0	3,083 [2,002, 4,296]	4	1,963	0.0	2,530 [2,016, 3,258]	5
steering DR, nominal	O-N-Hp	1,917	0.0	921 [461, 1,742]	4	1,963	0.0	1,669 [906, 2,472]	2
steering DR, online offset	O-N-Hp	1,917	0.0	869 [611, 1,306]	3	1,963	0.0	1,423 [964, 1,979]	2
steering DR, train calibrated	O-N-Hp	1,917	0.0	1,312 [409, 4,189]	4	1,963	0.0	1,651 [919, 2,430]	2
learned GRU (3 seeds)	O-N-Hp	1,917	0.0	543 [226, 2,478]	7	1,963	0.0	393 [305, 492]	5
learned physics residual (3 seeds)	O-N-Hp	1,917	0.0	690 [310, 2,994]	6	1,963	0.0	435 [366, 536]	4
map PF, speed only	O-M-Hp	1,917	0.0	286 [50, 868]	56	1,959	0.0	618 [186, 890]	48
map PF, steering	O-M-Hp	1,917	0.0	43 [29, 59]	87	1,959	0.0	27 [19, 47]	90
map PF, learned GRU motion (3 seeds)	O-M-Hp	1,917	0.0	89 [42, 168]	86	1,959	0.0	42 [29, 79]	86
map PF, learned residual motion (3 seeds)	O-M-Hp	1,917	0.0	87 [45, 262]	86	1,959	0.0	53 [33, 84]	86
MacroTracking	O-M-Hp	1,917	0.0	84 [66, 210]	62	1,959	0.0	52 [40, 143]	65
MacroTracking, train calibrated	O-M-Hp	1,917	0.0	76 [61, 172]	65	1,959	0.0	48 [38, 126]	67
OPD-II, online read-out	O-M-Hp	1,917	0.0	42 [27, 69]	79	1,959	0.0	24 [15, 54]	78
map beam, speed only	F-M-Hp	1,917	0.0	165 [37, 633]	54	1,959	0.0	343 [88, 637]	47
map beam, steering	F-M-Hp	1,917	0.0	37 [24, 54]	88	1,959	0.0	21 [17, 40]	92
map PF, steering, best-lineage read-out	F-M-Hp	1,917	0.0	39 [28, 56]	87	1,959	0.0	24 [17, 44]	90
Elastic Pathing (start heading used)	F-M-Hp	1,917	4.6	87 [38, 324]	58	1,959	4.0	80 [54, 165]	59
OPD-II (3 s delayed)	F-M-Hp	1,917	0.0	42 [27, 68]	79	1,959	0.0	24 [15, 54]	78
Dewri et al. (start heading used)	F-M-Hp	1,917	0.9	81 [38, 215]	62	1,959	2.8	89 [59, 157]	59
SPy, start known (start heading used)	F-M-Hp	1,917	0.3	1,474 [1,283, 1,602]	20	1,959	0.2	1,284 [1,090, 1,596]	22
map PF, steering	O-M-Hm	1,917	0.0	138 [99, 175]	79	1,959	0.0	88 [61, 124]	82
map PF, speed only	O-M-Hm	1,917	0.0	1,773 [1,550, 1,987]	34	1,959	0.0	1,868 [1,572, 2,097]	29
map beam, steering	F-M-Hm	1,917	0.0	41 [26, 59]	81	1,959	0.0	23 [17, 42]	87
map beam, speed only	F-M-Hm	1,917	0.0	1,314 [866, 1,619]	32	1,959	0.0	1,527 [1,294, 1,738]	29
Elastic Pathing	F-M-Hm	1,917	3.2	129 [42, 691]	53	1,959	2.1	177 [94, 344]	50
CAN-Trace (pedal subset), start known	F-M-Hm	245	99.2	fail [–, –]	0	162	96.9	fail [–, –]	0
CAN-Trace (pedal subset), 5 km area	F-M-Hm	245	65.3	fail [4,344, –]	0	162	60.5	fail [–, –]	0
Dewri et al.	F-M-Hm	1,917	0.4	293 [81, 775]	46	1,959	1.3	487 [272, 708]	43
SPy, start known	F-M-Hm	1,917	0.1	2,198 [1,702, 2,636]	11	1,959	0.1	1,810 [1,512, 2,211]	13
SPy, 5 km area	F-M-Hm	1,917	11.8	3,929 [3,222, 4,406]	0	1,959	5.5	3,400 [3,050, 3,953]	0
curvature DR (class F, not CAN-only)	O-N-Hp	1,917	0.0	84 [74, 107]	27	1,963	0.0	80 [70, 92]	27
map PF, curvature (class F)	O-M-Hp	1,917	0.0	44 [29, 61]	86	1,959	0.0	28 [19, 49]	89

Left block: test drivers; right block: few-shot drivers. Protocols O/F (online / offline), N/M (no map / map), Hp/Hm (start heading given / withheld).

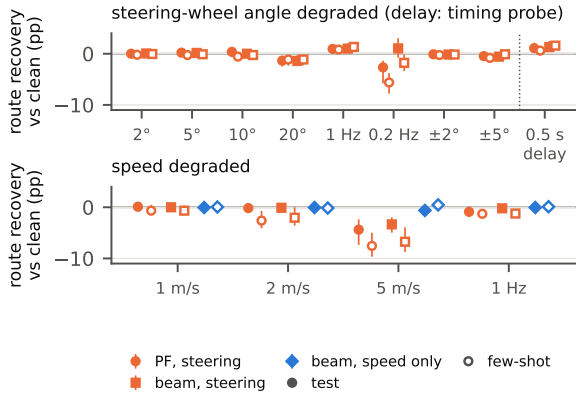


Fig. 2. Fixed map attacks on degraded inputs: change in the five-minute route recovery rate against the clean signal on the same windows (percentage points; 95 % driver-clustered CIs; filled: test, hollow: few-shot drivers); every level of the grid fixed before the first run; the 0.5 s delay is a timing probe.

TABLE III

COMPLETE BOARD, 15 MIN WINDOWS, BENCHMARK V1.1: EXPECTED WINDOWS n, FAILURE SHARE (NO OUTPUT), FAILURE-AWARE MEDIAN ADE IN METRES WITH THE 95 % DRIVER-CLUSTERED CI, AND THE ROUTE RECOVERY RATE (PATH RECALL AT 50 M ≥ 0.8), TEST AND FEW-SHOT DRIVERS.

method	protocol	n	fail %	median ADE m [CI]	route %	n	fail %	median ADE m [CI]	route %
hold start	O-N-Hp	608	0.0	8,759 [5,761, 12,194]	0	565	0.0	7,494 [5,973, 10,196]	0
straight line (speed)	O-N-Hp	608	0.0	3,389 [3,124, 4,037]	1	565	0.0	3,906 [3,630, 4,335]	1
start prior (speed, train stats)	O-N-Hp	608	0.0	3,601 [3,128, 3,841]	0	565	0.0	3,775 [3,545, 4,116]	0
route retrieval (speed, train routes)	O-N-Hp	608	0.0	8,365 [5,617, 12,123]	1	565	0.0	7,326 [5,648, 9,704]	3
steering DR, nominal	O-N-Hp	608	0.0	5,135 [2,276, 7,673]	1	565	0.0	7,174 [5,078, 9,162]	0
steering DR, online offset	O-N-Hp	608	0.0	4,805 [3,893, 6,281]	1	565	0.0	6,570 [5,331, 8,340]	0
steering DR, train calibrated	O-N-Hp	608	0.0	6,375 [2,069, 12,288]	1	565	0.0	7,138 [5,117, 9,181]	0
learned GRU (3 seeds)	O-N-Hp	608	0.0	3,010 [1,193, 10,461]	1	565	0.0	2,569 [1,923, 3,128]	0
learned physics residual (3 seeds)	O-N-Hp	608	0.0	4,799 [1,785, 10,746]	2	565	0.0	2,834 [2,292, 3,635]	0
map PF, speed only	O-M-Hp	608	0.0	2,326 [364, 4,401]	37	561	0.0	3,444 [2,231, 4,825]	27
map PF, steering	O-M-Hp	608	0.0	178 [85, 373]	69	561	0.0	127 [52, 224]	73
MacroTracking	O-M-Hp	608	0.0	851 [293, 3,183]	40	561	0.0	1,064 [245, 4,190]	39
MacroTracking, train calibrated	O-M-Hp	608	0.0	779 [284, 2,875]	39	561	0.0	1,110 [155, 3,912]	40
OPD-II, online read-out	O-M-Hp	608	0.0	223 [100, 963]	55	561	0.0	497 [94, 2,073]	47
map beam, speed only	F-M-Hp	608	0.0	2,142 [136, 4,651]	40	561	0.0	3,084 [1,498, 4,609]	30
map beam, steering	F-M-Hp	608	0.0	149 [68, 331]	73	561	0.0	106 [45, 174]	76
map PF, steering, best-lineage read-out	F-M-Hp	608	0.0	165 [79, 394]	68	561	0.0	125 [46, 215]	72
Elastic Pathing (start heading used)	F-M-Hp	608	9.2	1,094 [124, 3,263]	42	561	8.4	1,223 [928, 1,861]	42
OPD-II (3 s delayed)	F-M-Hp	608	0.0	223 [100, 962]	55	561	0.0	497 [94, 2,072]	47
Dewri et al. (start heading used)	F-M-Hp	608	5.3	1,053 [142, 2,985]	44	561	9.1	2,058 [1,159, 2,764]	37
SPy, start known (start heading used)	F-M-Hp	608	0.5	5,396 [4,436, 6,077]	6	561	0.5	4,819 [4,176, 6,440]	6
map PF, steering	O-M-Hm	608	0.0	340 [201, 706]	63	561	0.0	242 [118, 481]	68
map PF, speed only	O-M-Hm	608	0.0	5,799 [4,754, 6,909]	15	561	0.0	5,901 [4,652, 6,912]	10
map beam, steering	F-M-Hm	608	0.0	155 [68, 376]	70	561	0.0	110 [46, 184]	74
map beam, speed only	F-M-Hm	608	0.0	5,099 [3,859, 6,250]	25	561	0.0	5,419 [3,588, 6,052]	21
Elastic Pathing	F-M-Hm	608	6.6	1,214 [135, 3,777]	42	561	5.3	1,602 [1,001, 2,136]	40
CAN-Trace (pedal subset), 5 km area	F-M-Hm	79	93.7	fail [–, –]	0	45	95.6	fail [–, –]	0
Dewri et al.	F-M-Hm	608	2.5	3,064 [1,642, 4,239]	32	561	4.8	3,700 [2,838, 4,731]	27
SPy, start known	F-M-Hm	608	0.2	6,199 [4,874, 7,741]	4	561	0.4	5,928 [4,883, 7,689]	4
SPy, 5 km area	F-M-Hm	608	6.6	9,871 [6,632, 12,216]	0	561	4.8	8,349 [6,967, 10,942]	0
curvature DR (class F, not CAN-only)	O-N-Hp	608	0.0	511 [445, 644]	4	565	0.0	491 [424, 595]	2
map PF, curvature (class F)	O-M-Hp	608	0.0	186 [86, 430]	67	561	0.0	138 [61, 226]	71

Left block: test drivers; right block: few-shot drivers. Protocols O/F (online / offline), N/M (no map / map), Hp/Hm (start heading given / withheld).

TABLE IV

COMPLETE BOARD, WHOLE DRIVE WINDOWS, BENCHMARK V1.1: EXPECTED WINDOWS n, FAILURE SHARE (NO OUTPUT), FAILURE-AWARE MEDIAN ADE IN METRES WITH THE 95 % DRIVER-CLUSTERED CI, AND THE ROUTE RECOVERY RATE (PATH RECALL AT 50 M ≥ 0.8), TEST AND FEW-SHOT DRIVERS.

method	protocol	n	fail %	median ADE m [CI]	route %	n	fail %	median ADE m [CI]	route %
hold start	O-N-Hp	282	0.0	5,235 [3,825, 7,854]	0	460	0.0	3,322 [2,580, 4,870]	0
straight line (speed)	O-N-Hp	282	0.0	6,551 [4,548, 9,073]	2	460	0.0	3,327 [2,389, 4,744]	8
start prior (speed, train stats)	O-N-Hp	282	0.0	5,771 [3,750, 7,481]	2	460	0.0	2,912 [2,054, 3,825]	6
route retrieval (speed, train routes)	O-N-Hp	282	0.0	5,586 [3,843, 7,854]	1	460	0.0	3,236 [2,360, 4,800]	3
steering DR, nominal	O-N-Hp	282	0.0	2,708 [1,468, 8,341]	5	460	0.0	2,674 [1,251, 3,916]	7
steering DR, online offset	O-N-Hp	282	0.0	5,004 [3,266, 6,809]	4	460	0.0	3,118 [1,868, 4,449]	8
steering DR, train calibrated	O-N-Hp	282	0.0	3,121 [1,917, 8,403]	4	460	0.0	2,656 [1,498, 3,902]	7
learned GRU (3 seeds)	O-N-Hp	282	0.0	1,704 [1,149, 4,563]	6	460	0.0	887 [562, 1,547]	11
learned physics residual (3 seeds)	O-N-Hp	282	0.0	1,944 [1,255, 5,372]	6	460	0.0	1,100 [559, 1,838]	10
map PF, speed only	O-M-Hp	282	0.0	5,827 [3,211, 7,147]	8	454	0.0	2,908 [1,996, 3,946]	13
map PF, steering	O-M-Hp	282	0.0	1,906 [799, 3,128]	33	454	0.0	193 [91, 741]	57
MacroTracking	O-M-Hp	282	0.0	1,649 [779, 4,479]	33	454	0.0	227 [73, 900]	46
OPD-II, online read-out	O-M-Hp	282	0.0	1,712 [913, 4,120]	35	454	0.0	407 [52, 1,231]	50
map beam, speed only	F-M-Hp	282	0.0	5,930 [4,134, 8,365]	8	454	0.0	3,095 [2,044, 3,941]	12
map beam, steering	F-M-Hp	282	0.0	1,179 [343, 1,729]	41	454	0.0	94 [44, 239]	61
Elastic Pathing (start heading used)	F-M-Hp	282	11.0	3,597 [1,952, 5,600]	17	454	10.1	1,430 [957, 2,343]	29
OPD-II (3 s delayed)	F-M-Hp	282	0.0	1,711 [912, 4,120]	35	454	0.0	407 [53, 1,231]	50
Dewri et al. (start heading used)	F-M-Hp	282	13.1	5,382 [2,969, 7,392]	13	454	15.0	1,758 [1,119, 3,596]	22
SPy, start known (start heading used)	F-M-Hp	282	1.4	4,752 [3,473, 7,599]	4	454	0.9	2,651 [1,830, 4,240]	6
map PF, steering	O-M-Hm	282	0.0	2,239 [1,076, 3,319]	31	454	0.0	432 [230, 938]	51
map PF, speed only	O-M-Hm	282	0.0	6,313 [3,834, 8,070]	4	454	0.0	3,530 [2,312, 4,912]	9
map beam, steering	F-M-Hm	282	0.0	1,244 [447, 2,085]	39	454	0.0	124 [48, 510]	58
map beam, speed only	F-M-Hm	282	0.0	6,958 [3,931, 9,102]	6	454	0.0	3,605 [2,503, 4,779]	9
Elastic Pathing	F-M-Hm	282	6.7	3,554 [2,359, 5,631]	15	454	7.9	1,644 [1,056, 2,585]	25
Dewri et al.	F-M-Hm	282	8.9	5,619 [3,273, 7,780]	11	454	10.6	2,172 [1,429, 3,600]	19
SPy, start known	F-M-Hm	282	1.8	4,932 [3,613, 7,779]	2	454	1.1	3,254 [2,245, 4,387]	3
curvature DR (class F, not CAN-only)	O-N-Hp	282	0.0	678 [498, 1,427]	12	460	0.0	325 [204, 519]	21

Left block: test drivers; right block: few-shot drivers. Protocols O/F (online / offline), N/M (no map / map), Hp/Hm (start heading given / withheld).

TABLE V

SAME-COHORT HORIZON STUDY: ONE START PER DRIVE, SCORED 1, 5 AND 15 MINUTES AFTER THE START (MEDIAN ADE IN METRES / ROUTE RECOVERY %). ONLINE METHODS ON PREFIXES OF THEIR 15-MINUTE RUNS; OFFLINE METHODS RERUN ON EACH PREFIX.

method	protocol	1 min	5 min	15 min	1 min	5 min	15 min
dr_curvature	O-N-Hp	13 / 81	101 / 18	569 / 4	13 / 91	105 / 19	577 / 3
dr_none	O-N-Hm	382 / 4	2,150 / 0	7,693 / 0	439 / 7	2,749 / 0	8,806 / 0
dr_none	O-N-Hp	103 / 28	1,364 / 1	5,317 / 0	82 / 31	1,141 / 5	5,131 / 1
dr_steer	O-N-Hm	399 / 6	2,087 / 0	6,610 / 0	415 / 6	2,019 / 0	5,949 / 0
dr_steer	O-N-Hp	24 / 60	500 / 4	3,308 / 0	40 / 48	941 / 3	5,066 / 0
dr_steer_offonline	O-N-Hm	371 / 5	1,904 / 0	5,618 / 0	425 / 5	2,108 / 0	6,367 / 0
dr_steer_offonline	O-N-Hp	36 / 46	1,080 / 2	5,066 / 0	37 / 45	1,174 / 1	5,548 / 0
learned_gru_s0	O-N-Hm	385 / 7	2,189 / 0	7,032 / 0	426 / 6	2,331 / 0	6,733 / 0
learned_gru_s0	O-N-Hp	20 / 66	282 / 10	1,483 / 2	25 / 61	301 / 4	1,989 / 0
learned_gru_s1	O-N-Hm	388 / 7	2,255 / 0	6,927 / 0	424 / 7	2,361 / 1	6,740 / 0
learned_gru_s1	O-N-Hp	18 / 67	234 / 7	1,239 / 1	19 / 66	289 / 6	1,896 / 0
learned_gru_s2	O-N-Hm	384 / 6	2,207 / 0	7,107 / 0	425 / 7	2,260 / 0	6,884 / 0
learned_gru_s2	O-N-Hp	19 / 68	300 / 7	1,438 / 2	21 / 67	308 / 5	2,008 / 0
macrotracking	O-M-Hp	20 / 83	62 / 59	1,351 / 35	20 / 93	46 / 62	953 / 37
map_pf_curvature	O-M-Hm	94 / 72	182 / 61	707 / 47	64 / 78	95 / 69	662 / 51
map_pf_curvature	O-M-Hp	19 / 81	65 / 63	708 / 47	16 / 93	35 / 75	232 / 56
map_pf_none	O-M-Hm	271 / 44	1,309 / 18	5,180 / 7	278 / 44	1,506 / 24	5,191 / 7
map_pf_none	O-M-Hp	113 / 56	1,145 / 21	4,720 / 12	92 / 60	1,037 / 31	4,383 / 12
map_pf_steer	O-M-Hm	90 / 74	205 / 60	815 / 47	51 / 82	75 / 72	483 / 54
map_pf_steer	O-M-Hp	19 / 82	56 / 64	672 / 49	15 / 93	31 / 77	249 / 56
opd2_online	O-M-Hp	11 / 88	48 / 61	1,316 / 39	9 / 92	27 / 74	867 / 39
elastic_pathing	F-M-Hm	70 / 49	703 / 34	3,335 / 20	29 / 55	316 / 41	2,379 / 28
elastic_pathing_hp	F-M-Hp	23 / 59	505 / 37	3,437 / 20	14 / 72	238 / 42	2,267 / 29
map_beam_none	F-M-Hm	167 / 31	1,274 / 14	5,352 / 11	212 / 34	1,358 / 27	4,763 / 11
map_beam_none	F-M-Hp	97 / 41	1,030 / 18	5,107 / 14	56 / 51	941 / 33	3,798 / 13
map_beam_steer	F-M-Hm	21 / 70	44 / 71	297 / 58	14 / 79	27 / 79	126 / 62
map_beam_steer	F-M-Hp	17 / 81	44 / 70	303 / 58	12 / 94	25 / 84	123 / 63
map_pf_steer_lineage	F-M-Hp	17 / 81	53 / 66	675 / 47	13 / 93	27 / 77	241 / 55
opd2	F-M-Hp	11 / 88	48 / 62	1,316 / 38	9 / 91	26 / 73	867 / 39
can_trace_area	F-M-Hm	–	–	fail / 0	–	–	fail / 0
dewri	F-M-Hm	–	–	3,312 / 20	–	–	3,700 / 18
dewri_hp	F-M-Hp	–	–	3,069 / 25	–	–	3,562 / 22
dr_can_yaw	O-N-Hm	–	–	fail / 0	–	–	fail / 0
dr_can_yaw	O-N-Hp	–	–	fail / 0	–	–	fail / 0
dr_imu_yaw_rls	O-N-Hm	–	–	fail / 0	–	–	fail / 0
dr_imu_yaw_rls	O-N-Hp	–	–	fail / 0	–	–	fail / 0
dr_none_pn05	O-N-Hm	–	–	7,695 / 0	–	–	8,806 / 0
dr_none_pn05	O-N-Hp	–	–	5,318 / 0	–	–	5,145 / 1
dr_none_pq1	O-N-Hm	–	–	7,694 / 0	–	–	8,816 / 0
dr_none_pq1	O-N-Hp	–	–	5,306 / 0	–	–	5,116 / 1
dr_none_s1hz	O-N-Hm	–	–	7,693 / 0	–	–	8,806 / 0
dr_none_s1hz	O-N-Hp	–	–	5,317 / 0	–	–	5,131 / 1
dr_steer_calib	O-N-Hm	–	–	6,987 / 0	–	–	6,116 / 0
dr_steer_calib	O-N-Hp	–	–	3,138 / 1	–	–	5,052 / 0
dr_steer_pn05	O-N-Hm	–	–	6,605 / 0	–	–	5,980 / 0
dr_steer_pn05	O-N-Hp	–	–	3,286 / 1	–	–	5,047 / 0
dr_steer_pq1	O-N-Hm	–	–	6,653 / 0	–	–	5,969 / 0
dr_steer_pq1	O-N-Hp	–	–	3,473 / 0	–	–	5,069 / 0
dr_steer_s1hz	O-N-Hm	–	–	6,755 / 0	–	–	6,120 / 0
dr_steer_s1hz	O-N-Hp	–	–	3,278 / 2	–	–	4,939 / 0
learned_resid_nogru_s0	O-N-Hm	–	–	7,632 / 0	–	–	6,674 / 0
learned_resid_nogru_s0	O-N-Hp	–	–	3,898 / 0	–	–	4,718 / 0
learned_resid_nogru_s1	O-N-Hm	–	–	7,525 / 0	–	–	6,882 / 0
learned_resid_nogru_s1	O-N-Hp	–	–	3,792 / 0	–	–	5,024 / 0
learned_resid_nogru_s2	O-N-Hm	–	–	7,774 / 0	–	–	6,494 / 0
learned_resid_nogru_s2	O-N-Hp	–	–	3,719 / 0	–	–	4,664 / 0
learned_resid_s0	O-N-Hm	–	–	6,780 / 0	–	–	6,760 / 0
learned_resid_s0	O-N-Hp	–	–	2,253 / 2	–	–	2,285 / 0
learned_resid_s1	O-N-Hm	–	–	7,090 / 0	–	–	6,557 / 0
learned_resid_s1	O-N-Hp	–	–	1,910 / 2	–	–	2,050 / 0
learned_resid_s2	O-N-Hm	–	–	6,743 / 0	–	–	6,903 / 0
learned_resid_s2	O-N-Hp	–	–	1,803 / 2	–	–	2,582 / 0
macrotracking_calib	O-M-Hp	–	–	1,076 / 35	–	–	985 / 39
macrotracking_wrapfix	O-M-Hp	–	–	869 / 36	–	–	803 / 38
opd2_calib	F-M-Hp	–	–	1,175 / 41	–	–	970 / 41
opd2_wrapfix	F-M-Hp	–	–	1,316 / 38	–	–	867 / 39
p0_hold	O-N-Hm	–	–	5,432 / 0	–	–	5,512 / 0
p0_hold	O-N-Hp	–	–	5,432 / 0	–	–	5,512 / 0
prior	O-N-Hm	–	–	5,432 / 0	–	–	5,512 / 0
prior	O-N-Hp	–	–	4,391 / 0	–	–	4,328 / 0
retrieval	O-N-Hm	–	–	5,270 / 0	–	–	5,509 / 2
retrieval	O-N-Hp	–	–	5,270 / 0	–	–	5,509 / 2
spy	F-M-Hm	–	–	5,110 / 1	–	–	4,900 / 2
spy_area	F-M-Hm	–	–	5,917 / 0	–	–	5,765 / 0
spy_hp	F-M-Hp	–	–	4,761 / 2	–	–	4,592 / 2

Left: test drives; right: few-shot drives.

TABLE VI

SIGNAL STUDY (PEDAL-VALID FIVE-MINUTE WINDOWS OF THE HELD-OUT DRIVES): ONE LEARNER, ONE DECODER, PER SIGNAL SET AND SEED; FAILURE-AWARE MEDIAN ADE WITH 95 % DRIVER-CLUSTERED CI, ROUTE RECOVERY AND 100 M COVERAGE.

setting	set	seed	n	fail	median ADE m [CI]	route %	c100 %
no map	S1	0	407	0	1,079 [974, 1,209]	1	19
no map	S1	1	407	0	1,182 [1,005, 1,413]	1	18
no map	S1	2	407	0	1,128 [1,002, 1,324]	1	19
no map	S2c	0	407	0	1,156 [1,006, 1,383]	1	19
no map	S2c	1	407	0	1,156 [1,043, 1,323]	0	18
no map	S2c	2	407	0	1,028 [854, 1,320]	2	21
no map	S3	0	407	0	960 [289, 1,437]	7	29
no map	S3	1	407	0	375 [226, 639]	6	36
no map	S3	2	407	0	376 [238, 728]	6	35
no map	S4c	0	407	0	922 [312, 1,242]	5	29
no map	S4c	1	407	0	401 [232, 759]	8	38
no map	S4c	2	407	0	378 [274, 595]	7	36
no map	S2b	0	407	0	1,119 [992, 1,369]	1	19
no map	S4b	0	407	0	768 [264, 1,182]	5	30
map	S1	0	407	0	284 [169, 548]	50	46
map	S1	1	407	0	421 [332, 517]	48	39
map	S1	2	407	0	312 [201, 505]	51	39
map	S2c	0	407	0	367 [290, 507]	50	42
map	S2c	1	407	0	382 [294, 535]	50	43
map	S2c	2	407	0	490 [394, 702]	42	44
map	S3	0	407	0	48 [33, 61]	87	82
map	S3	1	407	0	39 [29, 67]	88	85
map	S3	2	407	0	58 [25, 73]	88	78
map	S4c	0	407	0	36 [28, 52]	86	86
map	S4c	1	407	0	42 [36, 62]	88	85
map	S4c	2	407	0	47 [34, 60]	89	84
map	S2b	0	407	0	351 [246, 513]	51	38
map	S4b	0	407	0	37 [28, 57]	87	86

TABLE VII

SIGNAL STUDY, PAIRED GAINS OF ADDING A SIGNAL (A − B, METRES; NEGATIVE = BETTER): MEDIAN AND MEAN PAIRED DIFFERENCES WITH 95 % CIS AND THE SHARE OF WINDOWS THAT IMPROVE.

setting	added	a	b	seed	n	median diff [CI]	mean diff [CI]	improved %
no map	pedals — speed	S2c	S1	0	407	42.9 [14.7, 70.6]	248 [107, 406]	41
no map	pedals — speed	S2c	S1	1	407	-2.3 [-42.8, 54.0]	-159 [-257, 27]	50
no map	pedals — speed	S2c	S1	2	407	-4.1 [-36.9, 20.1]	-81 [-233, 28]	51
no map	steering — speed	S3	S1	0	407	-134.5 [-515.2, 130.2]	25 [-614, 358]	56
no map	steering — speed	S3	S1	1	407	-733.9 [-905.2, -352.2]	-853 [-1,110, -359]	81
no map	steering — speed	S3	S1	2	407	-530.6 [-747.3, -251.7]	-734 [-963, -288]	75
no map	pedals — speed+steering	S4c	S3	0	407	-25.2 [-58.1, 14.7]	-85 [-135, 15]	58
no map	pedals — speed+steering	S4c	S3	1	407	-3.7 [-22.3, 36.8]	-28 [-177, 110]	52
no map	pedals — speed+steering	S4c	S3	2	407	-12.5 [-67.7, 45.8]	-58 [-219, 64]	54
no map	steering — speed+pedals	S4c	S2c	0	407	-258.1 [-658.4, -93.8]	-308 [-888, -21]	63
no map	steering — speed+pedals	S4c	S2c	1	407	-587.1 [-720.9, -419.4]	-722 [-898, -365]	83
no map	steering — speed+pedals	S4c	S2c	2	407	-405.2 [-652.0, -224.1]	-711 [-871, -413]	75
no map	pressed flags — speed	S2b	S1	0	407	76.2 [32.6, 102.2]	162 [83, 279]	40
no map	pressed flags — speed+steering	S4b	S3	0	407	-93.1 [-271.2, 5.0]	-218 [-325, -35]	70
map	pedals — speed	S2c	S1	0	407	0.9 [-1.6, 7.9]	13 [-22, 40]	48
map	pedals — speed	S2c	S1	1	407	-4.8 [-9.4, 0.6]	-2 [-48, 104]	56
map	pedals — speed	S2c	S1	2	407	6.1 [-6.6, 19.7]	150 [-2, 235]	44
map	steering — speed	S3	S1	0	407	-119.5 [-305.3, -80.9]	-698 [-893, -581]	83
map	steering — speed	S3	S1	1	407	-233.7 [-363.5, -149.9]	-727 [-856, -582]	87
map	steering — speed	S3	S1	2	407	-147.0 [-343.4, -103.8]	-690 [-895, -582]	88
map	pedals — speed+steering	S4c	S3	0	407	-6.4 [-10.9, -0.3]	-13 [-29, 20]	72
map	pedals — speed+steering	S4c	S3	1	407	0.9 [-1.2, 3.7]	16 [-14, 55]	44
map	pedals — speed+steering	S4c	S3	2	407	-3.7 [-9.0, 3.8]	-14 [-40, 8]	63
map	steering — speed+pedals	S4c	S2c	0	407	-191.3 [-320.7, -144.2]	-724 [-890, -583]	90
map	steering — speed+pedals	S4c	S2c	1	407	-231.4 [-348.8, -127.1]	-709 [-894, -570]	84
map	steering — speed+pedals	S4c	S2c	2	407	-359.2 [-461.9, -207.2]	-855 [-1,030, -621]	85
map	pressed flags — speed	S2b	S1	0	407	5.2 [0.8, 10.7]	14 [-20, 77]	41
map	pressed flags — speed+steering	S4b	S3	0	407	-3.6 [-12.2, 0.4]	-30 [-59, 2]	67

TABLE VIII

CANDIDATE SETS ON FIVE-MINUTE WINDOWS: WHOLE-PATH ORACLE MEDIAN ADE AMONG THE FIRST K CANDIDATES, SHARE OF WINDOWS WHOSE BEST CANDIDATE IS RANKED FIRST, AND THE MEDIAN NUMBER OF CANDIDATES.

method	protocol	fold	n	out	single	K=1	3	5	10	best 1st %	cands
map_beam_steer	F-M-Hp	test	1,917	1,917	31	37	30	28	27	58	10
map_beam_steer	F-M-Hp	few-shot	1,959	1,959	15	21	18	18	17	61	10
map_beam_steer	F-M-Hm	test	1,917	1,917	2	41	32	29	27	54	10
map_beam_steer	F-M-Hm	few-shot	1,959	1,959	0	23	19	18	17	58	10
map_beam_none	F-M-Hp	test	1,917	1,917	31	165	85	73	60	35	10
map_beam_none	F-M-Hp	few-shot	1,959	1,959	18	343	89	73	56	28	10
map_beam_none	F-M-Hm	test	1,917	1,917	2	1,314	210	94	72	21	10
map_beam_none	F-M-Hm	few-shot	1,959	1,959	0	1,527	461	139	71	17	10
elastic_pathing_hp	F-M-Hp	test	1,917	1,828	1,402	87	77	76	75	84	1
elastic_pathing_hp	F-M-Hp	few-shot	1,959	1,881	1,488	80	66	65	64	85	1
elastic_pathing	F-M-Hm	test	1,917	1,855	1,484	129	103	98	96	87	1
elastic_pathing	F-M-Hm	few-shot	1,959	1,918	1,577	177	114	111	111	87	1
can_trace	F-M-Hm	test	245	2	0	fail	fail	fail	fail	50	2
can_trace	F-M-Hm	few-shot	162	5	2	fail	fail	fail	fail	80	2
can_trace_area	F-M-Hm	test	245	85	14	fail	fail	fail	fail	36	6
can_trace_area	F-M-Hm	few-shot	162	64	5	fail	fail	fail	fail	27	6
dewri_hp	F-M-Hp	test	1,917	1,899	347	81	58	51	45	50	9
dewri_hp	F-M-Hp	few-shot	1,959	1,905	311	89	49	41	35	47	6
dewri	F-M-Hm	test	1,917	1,909	40	293	70	60	49	39	10
dewri	F-M-Hm	few-shot	1,959	1,933	40	487	65	48	38	34	10

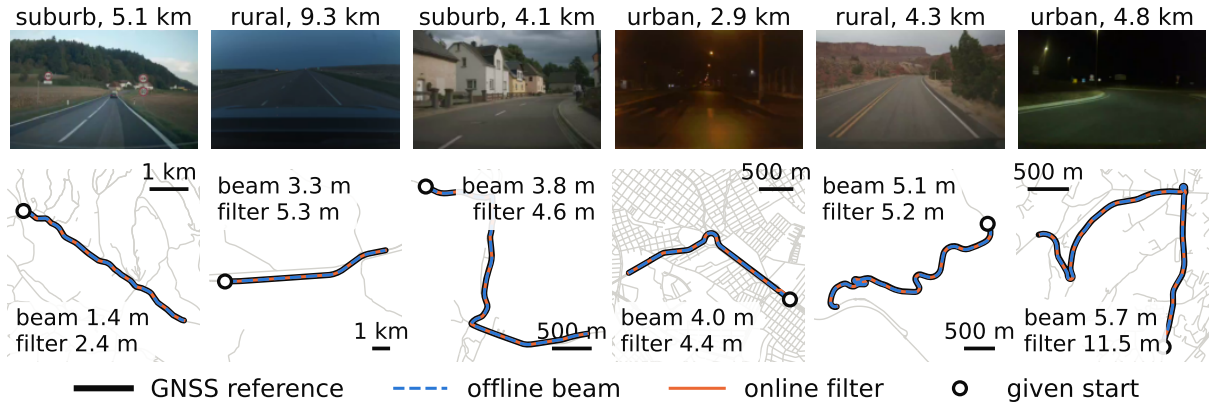


Fig. 3. Six held-out five-minute windows recovered by the offline beam and the online filter (speed, steering, start and heading known) below their camera frames at mid-window (not an input); grey: roads; numbers: ADE.

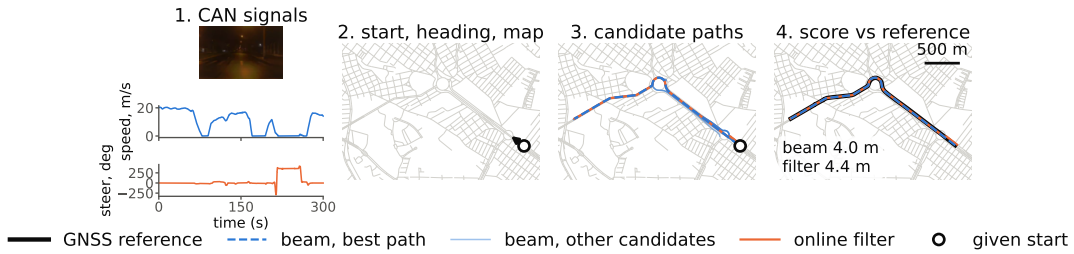


Fig. 4. The pipeline on one held-out window: CAN signals, start, heading and visible roads in; candidate paths; scoring at the reference seconds.

TABLE IX

FIXED MAP ATTACKS ON DEGRADED INPUTS (FIVE-MINUTE WINDOWS, GRID FIXED BEFORE THE FIRST RUN): MEDIAN ADE AND ROUTE RECOVERY ON THE CLEAN AND THE DEGRADED SIGNAL, AND THE PAIRED CHANGE OF THE ROUTE RECOVERY RATE IN PERCENTAGE POINTS WITH 95 % DRIVER-CLUSTERED CI.

method	protocol	degradation	fold	n	med. clean	med. degraded	route clean %	route degraded %	Δ route pp [CI]
map PF, steering	O-M-Hp	steering quantised to 2 deg	test	1,917	43	43	87	87	+0.0 [-0.2, +0.3]
map PF, steering	O-M-Hp	steering quantised to 2 deg	few-shot	1,959	27	27	90	90	-0.2 [-0.6, +0.0]
map PF, steering	O-M-Hp	steering quantised to 5 deg	test	1,917	43	42	87	87	+0.2 [-0.2, +0.7]
map PF, steering	O-M-Hp	steering quantised to 5 deg	few-shot	1,959	27	26	90	90	-0.3 [-0.8, +0.1]
map PF, steering	O-M-Hp	steering quantised to 10 deg	test	1,917	43	42	87	87	+0.4 [-0.1, +0.9]
map PF, steering	O-M-Hp	steering quantised to 10 deg	few-shot	1,959	27	27	90	90	-0.6 [-1.3, -0.0]
map PF, steering	O-M-Hp	steering quantised to 20 deg	test	1,917	43	44	87	86	-1.4 [-2.5, -0.4]
map PF, steering	O-M-Hp	steering quantised to 20 deg	few-shot	1,959	27	28	90	89	-1.1 [-2.3, -0.4]
map PF, steering	O-M-Hp	steering at 1 Hz	test	1,917	43	42	87	88	+0.9 [+0.4, +1.8]
map PF, steering	O-M-Hp	steering at 1 Hz	few-shot	1,959	27	27	90	91	+0.8 [+0.3, +1.3]
map PF, steering	O-M-Hp	steering at 0.2 Hz	test	1,917	43	49	87	84	-2.7 [-5.8, -1.4]
map PF, steering	O-M-Hp	steering at 0.2 Hz	few-shot	1,959	27	38	90	85	-5.6 [-7.8, -3.8]
map PF, steering	O-M-Hp	steering offset noise 2 deg	test	1,917	43	42	87	87	-0.1 [-0.8, +0.3]
map PF, steering	O-M-Hp	steering offset noise 2 deg	few-shot	1,959	27	27	90	90	-0.3 [-0.8, +0.2]
map PF, steering	O-M-Hp	steering offset noise 5 deg	test	1,917	43	43	87	87	-0.5 [-1.2, +0.0]
map PF, steering	O-M-Hp	steering offset noise 5 deg	few-shot	1,959	27	27	90	89	-0.8 [-1.7, -0.1]
map PF, steering	O-M-Hp	steering delayed 0.5 s	test	1,917	43	42	87	88	+1.1 [+0.4, +2.1]
map PF, steering	O-M-Hp	steering delayed 0.5 s	few-shot	1,959	27	27	90	91	+0.6 [-0.3, +1.2]
map PF, steering	O-M-Hp	speed quantised to 1 m/s	test	1,917	43	42	87	87	+0.1 [-0.5, +0.5]
map PF, steering	O-M-Hp	speed quantised to 1 m/s	few-shot	1,959	27	31	90	89	-0.7 [-1.4, +0.5]
map PF, steering	O-M-Hp	speed quantised to 2 m/s	test	1,917	43	41	87	87	-0.2 [-0.7, +0.6]
map PF, steering	O-M-Hp	speed quantised to 2 m/s	few-shot	1,959	27	42	90	88	-2.6 [-4.1, -0.7]
map PF, steering	O-M-Hp	speed quantised to 5 m/s	test	1,917	43	122	87	83	-4.4 [-7.3, -2.4]
map PF, steering	O-M-Hp	speed quantised to 5 m/s	few-shot	1,959	27	68	90	83	-7.6 [-9.7, -5.0]
map PF, steering	O-M-Hp	speed at 1 Hz	test	1,917	43	44	87	86	-0.9 [-1.9, -0.1]
map PF, steering	O-M-Hp	speed at 1 Hz	few-shot	1,959	27	29	90	89	-1.3 [-1.8, -0.5]
map beam, steering	F-M-Hp	steering quantised to 2 deg	test	1,917	37	37	88	88	+0.1 [+0.0, +0.2]
map beam, steering	F-M-Hp	steering quantised to 2 deg	few-shot	1,959	21	21	92	92	-0.1 [-0.2, +0.0]
map beam, steering	F-M-Hp	steering quantised to 5 deg	test	1,917	37	36	88	88	+0.2 [-0.1, +0.8]
map beam, steering	F-M-Hp	steering quantised to 5 deg	few-shot	1,959	21	22	92	92	-0.1 [-0.4, +0.2]
map beam, steering	F-M-Hp	steering quantised to 10 deg	test	1,917	37	37	88	88	+0.0 [-0.3, +0.4]
map beam, steering	F-M-Hp	steering quantised to 10 deg	few-shot	1,959	21	22	92	92	-0.3 [-0.7, +0.1]
map beam, steering	F-M-Hp	steering quantised to 20 deg	test	1,917	37	39	88	86	-1.4 [-2.4, -0.3]
map beam, steering	F-M-Hp	steering quantised to 20 deg	few-shot	1,959	21	25	92	91	-1.1 [-2.2, -0.4]
map beam, steering	F-M-Hp	steering at 1 Hz	test	1,917	37	37	88	89	+1.0 [+0.2, +2.0]
map beam, steering	F-M-Hp	steering at 1 Hz	few-shot	1,959	21	22	92	93	+1.3 [+0.8, +2.0]
map beam, steering	F-M-Hp	steering at 0.2 Hz	test	1,917	37	46	88	89	+1.0 [-0.8, +3.0]
map beam, steering	F-M-Hp	steering at 0.2 Hz	few-shot	1,959	21	36	92	90	-1.8 [-3.4, -0.1]
map beam, steering	F-M-Hp	steering offset noise 2 deg	test	1,917	37	37	88	87	-0.2 [-0.4, +0.0]
map beam, steering	F-M-Hp	steering offset noise 2 deg	few-shot	1,959	21	22	92	92	-0.2 [-0.4, +0.2]
map beam, steering	F-M-Hp	steering offset noise 5 deg	test	1,917	37	37	88	87	-0.6 [-1.4, -0.1]
map beam, steering	F-M-Hp	steering offset noise 5 deg	few-shot	1,959	21	23	92	92	-0.1 [-0.5, +0.5]
map beam, steering	F-M-Hp	steering delayed 0.5 s	test	1,917	37	37	88	89	+1.3 [+0.5, +2.5]
map beam, steering	F-M-Hp	steering delayed 0.5 s	few-shot	1,959	21	22	92	94	+1.6 [+0.8, +2.4]
map beam, steering	F-M-Hp	speed quantised to 1 m/s	test	1,917	37	36	88	88	+0.0 [-0.7, +0.6]
map beam, steering	F-M-Hp	speed quantised to 1 m/s	few-shot	1,959	21	26	92	91	-0.7 [-1.2, +0.3]
map beam, steering	F-M-Hp	speed quantised to 2 m/s	test	1,917	37	37	88	87	-0.1 [-0.7, +0.8]
map beam, steering	F-M-Hp	speed quantised to 2 m/s	few-shot	1,959	21	34	92	90	-2.0 [-3.6, +0.0]
map beam, steering	F-M-Hp	speed quantised to 5 m/s	test	1,917	37	105	88	84	-3.3 [-5.0, -2.0]
map beam, steering	F-M-Hp	speed quantised to 5 m/s	few-shot	1,959	21	59	92	85	-6.7 [-8.7, -3.9]
map beam, steering	F-M-Hp	speed at 1 Hz	test	1,917	37	37	88	87	-0.2 [-1.1, +0.5]
map beam, steering	F-M-Hp	speed at 1 Hz	few-shot	1,959	21	23	92	91	-1.2 [-1.8, -0.6]
map beam, speed only	F-M-Hp	speed quantised to 1 m/s	test	1,917	165	169	54	54	-0.1 [-0.3, +0.1]
map beam, speed only	F-M-Hp	speed quantised to 1 m/s	few-shot	1,959	343	338	47	47	+0.1 [-0.4, +0.7]
map beam, speed only	F-M-Hp	speed quantised to 2 m/s	test	1,917	165	174	54	54	-0.1 [-0.5, +0.3]
map beam, speed only	F-M-Hp	speed quantised to 2 m/s	few-shot	1,959	343	336	47	47	-0.2 [-0.6, +0.4]
map beam, speed only	F-M-Hp	speed quantised to 5 m/s	test	1,917	165	247	54	53	-0.6 [-1.5, +0.1]
map beam, speed only	F-M-Hp	speed quantised to 5 m/s	few-shot	1,959	343	323	47	47	+0.5 [-0.5, +1.1]
map beam, speed only	F-M-Hp	speed at 1 Hz	test	1,917	165	169	54	54	-0.1 [-0.3, +0.1]
map beam, speed only	F-M-Hp	speed at 1 Hz	few-shot	1,959	343	343	47	47	+0.1 [+0.0, +0.3]

TABLE X

DEGRADED INPUTS FOR DEAD RECKONING (NO MAP): CLEAN AND DEGRADED MEDIANS AND THE PAIRED MEDIAN / MEAN DIFFERENCES (METRES).

method	degradation	fold	med. clean	med. degraded	n	paired median [CI]	paired mean [CI]
steering DR	speed quantised to 1 m/s	test	921	920	1,917	-0.1 [-0.3, 0.2]	0 [-1, 1]
steering DR	speed quantised to 1 m/s	few-shot	1,669	1,664	1,963	-0.1 [-0.6, 0.1]	-0 [-1, 1]
steering DR	speed noise 0.5 m/s	test	921	926	1,917	0.1 [-0.0, 0.2]	1 [0, 2]
steering DR	speed noise 0.5 m/s	few-shot	1,669	1,670	1,963	0.1 [0.0, 0.3]	2 [1, 3]
steering DR	steering at 1 Hz	test	921	934	1,917	0.4 [-1.3, 1.6]	7 [-1, 14]
steering DR	steering at 1 Hz	few-shot	1,669	1,683	1,963	0.2 [-1.0, 2.3]	5 [-1, 15]
straight line	speed quantised to 1 m/s	test	735	736	1,917	0.0 [-0.1, 0.2]	0 [0, 1]
straight line	speed quantised to 1 m/s	few-shot	825	824	1,963	0.1 [-0.1, 0.5]	1 [-0, 2]
straight line	speed noise 0.5 m/s	test	735	736	1,917	0.1 [0.0, 0.3]	1 [1, 2]
straight line	speed noise 0.5 m/s	few-shot	825	825	1,963	0.1 [0.1, 0.2]	1 [1, 1]
straight line	steering at 1 Hz	test	735	735	1,917	0.0 [0.0, 0.0]	0 [0, 0]
straight line	steering at 1 Hz	few-shot	825	825	1,963	0.0 [0.0, 0.0]	0 [0, 0]

TABLE XI

COST STUDY ON ONE FIXED VALIDATION SUBSET, ONE WORKER PER METHOD: INFERENCE SECONDS PER WINDOW (MEDIAN AND 90TH PERCENTILE; FAILED SEARCHES COUNT WITH THE TIME THEY RAN), PEAK MEMORY AND THE PER-DRIVE GRAPH BUILD.

method	n_5	fail	med s	p90 s	n_{15}	fail	med s	p90 s	RSS GB	graph s
dewri	76	0	0.04	0.09	16	1	0.33	0.83	1.8	2.7
dewri_hp	76	1	0.03	0.06	16	2	0.23	0.60	1.8	2.7
dr_none	76	0	0.02	0.02	16	0	0.06	0.06	0.8	–
dr_steer	76	0	0.03	0.03	16	0	0.09	0.09	0.8	–
elastic_pathing	76	0	0.01	0.13	16	0	0.15	0.54	1.8	2.6
elastic_pathing_hp	76	0	0.01	0.12	16	0	0.16	0.52	1.8	2.6
learned_gru	76	0	0.22	0.23	16	0	0.66	0.67	0.8	–
macrotracking	76	0	0.08	0.21	16	0	0.43	0.69	1.8	2.6
map_beam_none	76	0	0.12	0.18	16	0	0.43	0.50	1.8	2.6
map_beam_steer	76	0	0.24	0.51	16	0	0.93	1.33	1.8	2.6
map_beam_steer_b50	76	0	0.07	0.13	16	0	0.25	0.37	1.8	2.6
map_pf_learned	76	0	0.83	0.95	16	0	2.56	2.89	1.7	2.6
map_pf_none	76	0	0.89	1.38	16	0	4.03	4.97	1.7	2.6
map_pf_steer	76	0	0.57	0.66	16	0	1.77	2.05	1.7	2.6
map_pf_steer_n500	76	0	0.22	0.30	16	0	0.70	0.87	1.7	2.6
opd2	76	0	0.08	0.23	16	0	0.19	0.28	2.1	2.6
spy	76	0	0.13	0.68	16	0	0.24	1.23	1.8	2.7
spy_area	76	9	4.93	13.53	16	1	7.71	23.97	1.8	2.7
spy_hp	76	0	0.08	0.32	16	0	0.22	0.62	1.8	2.7

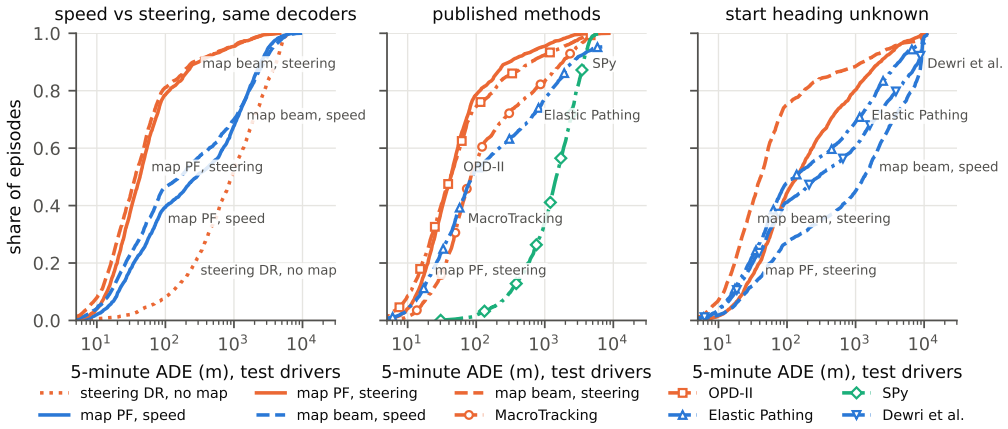


Fig. 5. ADE distributions of the main rows on five-minute windows.

TABLE XII

ACCURACY ON THE COST SUBSET AT THE PUBLISHED AND AT SMALLER SEARCH BUDGETS (VALIDATION WINDOWS).

method / budget	protocol	window	n	fail	median ADE m	route %
dewri__G1	F-M-Hm	5 min	76	0	1,233	21
dewri__G1	F-M-Hm	15 min	16	1	4,170	12
dewri_hp__G1	F-M-Hp	5 min	76	1	524	34
dewri_hp__G1	F-M-Hp	15 min	16	2	1,737	25
dr_none__G1	O-N-Hp	5 min	76	0	497	16
dr_none__G1	O-N-Hp	15 min	16	0	3,132	6
dr_steer__G3	O-N-Hp	5 min	76	0	745	3
dr_steer__G3	O-N-Hp	15 min	16	0	4,681	0
elastic_pathing__G1	F-M-Hm	5 min	76	0	428	38
elastic_pathing__G1	F-M-Hm	15 min	16	0	3,268	19
elastic_pathing_hp__G1	F-M-Hp	5 min	76	0	205	46
elastic_pathing_hp__G1	F-M-Hp	15 min	16	0	2,100	25
learned_gru_s0__G3	O-N-Hp	5 min	76	0	203	12
learned_gru_s0__G3	O-N-Hp	15 min	16	0	1,237	0
macrotracking__G3	O-M-Hp	5 min	76	0	75	50
macrotracking__G3	O-M-Hp	15 min	16	0	1,709	25
map_beam_none__G1	F-M-Hp	5 min	76	0	549	32
map_beam_none__G1	F-M-Hp	15 min	16	0	4,260	12
map_beam_steer__G3	F-M-Hp	5 min	76	0	21	84
map_beam_steer__G3	F-M-Hp	15 min	16	0	58	81
map_beam_steer_b50__G3	F-M-Hp	5 min	76	0	21	84
map_beam_steer_b50__G3	F-M-Hp	15 min	16	0	58	81
map_pf_learned_gru_s0__G3	O-M-Hp	5 min	76	0	38	84
map_pf_learned_gru_s0__G3	O-M-Hp	15 min	16	0	115	88
map_pf_none__G1	O-M-Hp	5 min	76	0	933	36
map_pf_none__G1	O-M-Hp	15 min	16	0	3,935	19
map_pf_steer__G3	O-M-Hp	5 min	76	0	26	87
map_pf_steer__G3	O-M-Hp	15 min	16	0	90	81
map_pf_steer_n500__G3	O-M-Hp	5 min	76	0	24	87
map_pf_steer_n500__G3	O-M-Hp	15 min	16	0	91	81
opd2__G3	F-M-Hp	5 min	76	0	23	76
opd2__G3	F-M-Hp	15 min	16	0	1,317	31
spy__G3	F-M-Hm	5 min	76	0	1,306	7
spy__G3	F-M-Hm	15 min	16	0	5,401	6
spy_area__G3	F-M-Hm	5 min	76	9	2,743	0
spy_area__G3	F-M-Hm	15 min	16	1	6,180	0
spy_hp__G3	F-M-Hp	5 min	76	0	965	16
spy_hp__G3	F-M-Hp	15 min	16	0	4,946	6

TABLE XIII

CANDIDATE CONFIDENCE SCORE OF THE OFFLINE BEAM: AUROC OF THE TOP-1 SCORE FOR ROUTE RECOVERY (DRIVER-CLUSTERED CI), SPEARMAN CORRELATION WITH THE ADE, AND THE SHARE OF WINDOWS WHOSE SCORE IS 1.

method	protocol	window	fold	n	fail	AUROC [CI]	Spearman	score = 1 %
map_beam_steer	F-M-Hp	5 min	test	1,917	0	0.47 [0.41, 0.52]	-0.01	29
map_beam_steer	F-M-Hp	5 min	few-shot	1,959	0	0.49 [0.44, 0.54]	-0.06	38
map_beam_steer	F-M-Hp	15 min	test	608	0	0.54 [0.48, 0.59]	-0.08	19
map_beam_steer	F-M-Hp	15 min	few-shot	561	0	0.54 [0.47, 0.59]	-0.20	34
map_beam_steer	F-M-Hm	5 min	test	1,917	0	0.57 [0.53, 0.62]	-0.11	16
map_beam_steer	F-M-Hm	5 min	few-shot	1,959	0	0.57 [0.51, 0.61]	-0.15	27
map_beam_steer	F-M-Hm	15 min	test	608	0	0.55 [0.50, 0.60]	-0.09	15
map_beam_steer	F-M-Hm	15 min	few-shot	561	0	0.58 [0.48, 0.65]	-0.26	30
map_beam_none	F-M-Hp	5 min	test	1,917	0	0.81 [0.73, 0.86]	-0.51	7
map_beam_none	F-M-Hp	5 min	few-shot	1,959	0	0.79 [0.74, 0.84]	-0.40	8
map_beam_none	F-M-Hp	15 min	test	608	0	0.79 [0.63, 0.85]	-0.42	3
map_beam_none	F-M-Hp	15 min	few-shot	561	0	0.77 [0.70, 0.82]	-0.26	2
map_beam_none	F-M-Hm	5 min	test	1,917	0	0.65 [0.59, 0.67]	0.02	1
map_beam_none	F-M-Hm	5 min	few-shot	1,959	0	0.68 [0.65, 0.71]	0.03	0
map_beam_none	F-M-Hm	15 min	test	608	0	0.69 [0.56, 0.71]	0.04	0
map_beam_none	F-M-Hm	15 min	few-shot	561	0	0.69 [0.62, 0.73]	0.07	0

TABLE XIV

SELECTIVE PREDICTION BY THE BEAM'S CONFIDENCE SCORE: WINDOWS ACCEPTED AT A THRESHOLD (COVERAGE, ALL EXPECTED WINDOWS IN THE DENOMINATOR), ROUTE RECOVERY AMONG THE ACCEPTED AND THE SHARE OF ACCEPTED ROUTES THAT ARE WRONG.

method	protocol	window	fold	τ	coverage %	route among accepted % [CI]	wrong %
map_beam_steer	F-M-Hp	5 min	test	0.5	77	86 [80, 92]	14
map_beam_steer	F-M-Hp	5 min	test	0.7	57	86 [79, 92]	14
map_beam_steer	F-M-Hp	5 min	test	0.9	46	87 [81, 92]	13
map_beam_steer	F-M-Hp	5 min	few-shot	0.5	85	92 [88, 94]	8
map_beam_steer	F-M-Hp	5 min	few-shot	0.7	65	91 [86, 94]	9
map_beam_steer	F-M-Hp	5 min	few-shot	0.9	55	92 [87, 94]	8
map_beam_steer	F-M-Hp	15 min	test	0.5	84	73 [59, 83]	27
map_beam_steer	F-M-Hp	15 min	test	0.7	61	74 [59, 85]	26
map_beam_steer	F-M-Hp	15 min	test	0.9	45	76 [61, 85]	24
map_beam_steer	F-M-Hp	15 min	few-shot	0.5	89	76 [70, 81]	24
map_beam_steer	F-M-Hp	15 min	few-shot	0.7	69	76 [68, 80]	24
map_beam_steer	F-M-Hp	15 min	few-shot	0.9	54	76 [70, 81]	24
map_beam_steer	F-M-Hm	5 min	test	0.5	70	83 [77, 88]	17
map_beam_steer	F-M-Hm	5 min	test	0.7	50	84 [78, 89]	16
map_beam_steer	F-M-Hm	5 min	test	0.9	36	85 [79, 91]	15
map_beam_steer	F-M-Hm	5 min	few-shot	0.5	80	87 [84, 90]	13
map_beam_steer	F-M-Hm	5 min	few-shot	0.7	61	88 [84, 90]	12
map_beam_steer	F-M-Hm	5 min	few-shot	0.9	48	89 [85, 92]	11
map_beam_steer	F-M-Hm	15 min	test	0.5	82	72 [58, 83]	28
map_beam_steer	F-M-Hm	15 min	test	0.7	58	73 [58, 84]	27
map_beam_steer	F-M-Hm	15 min	test	0.9	42	74 [58, 86]	26
map_beam_steer	F-M-Hm	15 min	few-shot	0.5	88	75 [70, 81]	25
map_beam_steer	F-M-Hm	15 min	few-shot	0.7	69	75 [68, 79]	25
map_beam_steer	F-M-Hm	15 min	few-shot	0.9	53	79 [72, 83]	21
map_beam_none	F-M-Hp	5 min	test	0.5	28	89 [72, 95]	11
map_beam_none	F-M-Hp	5 min	test	0.7	19	94 [84, 97]	6
map_beam_none	F-M-Hp	5 min	test	0.9	13	97 [89, 99]	3
map_beam_none	F-M-Hp	5 min	few-shot	0.5	25	82 [73, 90]	18
map_beam_none	F-M-Hp	5 min	few-shot	0.7	15	91 [83, 96]	9
map_beam_none	F-M-Hp	5 min	few-shot	0.9	11	94 [88, 98]	6
map_beam_none	F-M-Hp	15 min	test	0.5	20	86 [51, 95]	14
map_beam_none	F-M-Hp	15 min	test	0.7	13	91 [64, 96]	9
map_beam_none	F-M-Hp	15 min	test	0.9	7	93 [60, 99]	7
map_beam_none	F-M-Hp	15 min	few-shot	0.5	16	67 [42, 81]	33
map_beam_none	F-M-Hp	15 min	few-shot	0.7	8	84 [55, 94]	16
map_beam_none	F-M-Hp	15 min	few-shot	0.9	5	93 [76, 100]	7
map_beam_none	F-M-Hm	5 min	test	0.5	17	50 [39, 55]	50
map_beam_none	F-M-Hm	5 min	test	0.7	6	56 [37, 69]	44
map_beam_none	F-M-Hm	5 min	test	0.9	2	75 [40, 90]	25
map_beam_none	F-M-Hm	5 min	few-shot	0.5	14	46 [42, 53]	54
map_beam_none	F-M-Hm	5 min	few-shot	0.7	5	58 [51, 70]	42
map_beam_none	F-M-Hm	5 min	few-shot	0.9	2	62 [49, 78]	38
map_beam_none	F-M-Hm	15 min	test	0.5	18	50 [23, 58]	50
map_beam_none	F-M-Hm	15 min	test	0.7	9	58 [34, 71]	42
map_beam_none	F-M-Hm	15 min	test	0.9	2	57 [8, 94]	43
map_beam_none	F-M-Hm	15 min	few-shot	0.5	10	42 [26, 54]	58
map_beam_none	F-M-Hm	15 min	few-shot	0.7	5	50 [33, 70]	50
map_beam_none	F-M-Hm	15 min	few-shot	0.9	2	36 [12, 57]	64

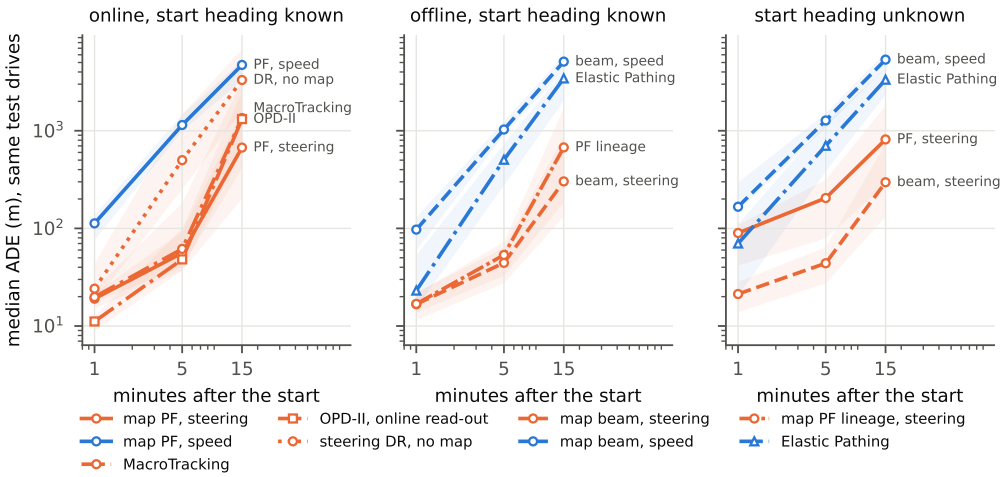


Fig. 6. Same-cohort horizon study: median ADE 1, 5 and 15 minutes after the start with driver-resampling bands.

TABLE XV

WINDOW TAXONOMY (FIVE MINUTES, WHOLE-PATH CRITERION AT THE GIVEN THRESHOLD): NO OUTPUT, NO CANDIDATE WITHIN THE THRESHOLD, A LOWER-RANKED CANDIDATE WITHIN IT WHILE THE TOP-1 IS NOT (RANKING FAILURE), TOP-1 WITHIN IT.

method	protocol	fold	thr. m	n	no output %	no cand. %	ranking %	Top-1 ok %
map_beam_steer	F-M-Hp	test	50	1,917	0	28	10	61
map_beam_steer	F-M-Hp	test	100	1,917	0	12	7	81
map_beam_steer	F-M-Hp	few-shot	50	1,959	0	19	7	74
map_beam_steer	F-M-Hp	few-shot	100	1,959	0	6	6	88
map_beam_steer	F-M-Hm	test	50	1,917	0	29	15	57
map_beam_steer	F-M-Hm	test	100	1,917	0	13	12	75
map_beam_steer	F-M-Hm	few-shot	50	1,959	0	19	11	70
map_beam_steer	F-M-Hm	few-shot	100	1,959	0	7	10	83
map_beam_none	F-M-Hp	test	50	1,917	0	55	13	32
map_beam_none	F-M-Hp	test	100	1,917	0	41	13	46
map_beam_none	F-M-Hp	few-shot	50	1,959	0	53	18	29
map_beam_none	F-M-Hp	few-shot	100	1,959	0	40	18	42
map_beam_none	F-M-Hm	test	50	1,917	0	58	24	18
map_beam_none	F-M-Hm	test	100	1,917	0	45	28	27
map_beam_none	F-M-Hm	few-shot	50	1,959	0	57	27	16
map_beam_none	F-M-Hm	few-shot	100	1,959	0	45	30	25
dewri_hp	F-M-Hp	test	50	1,917	1	45	18	37
dewri_hp	F-M-Hp	test	100	1,917	1	25	20	54
dewri_hp	F-M-Hp	few-shot	50	1,959	3	37	22	39
dewri_hp	F-M-Hp	few-shot	100	1,959	3	22	23	52
dewri	F-M-Hm	test	50	1,917	0	49	22	29
dewri	F-M-Hm	test	100	1,917	0	30	28	41
dewri	F-M-Hm	few-shot	50	1,959	1	41	29	29
dewri	F-M-Hm	few-shot	100	1,959	1	27	32	39
elastic_pathing_hp	F-M-Hp	test	50	1,917	5	56	4	35
elastic_pathing_hp	F-M-Hp	test	100	1,917	5	40	3	52
elastic_pathing_hp	F-M-Hp	few-shot	50	1,959	4	52	5	39
elastic_pathing_hp	F-M-Hp	few-shot	100	1,959	4	37	5	54
elastic_pathing	F-M-Hm	test	50	1,917	3	61	3	33
elastic_pathing	F-M-Hm	test	100	1,917	3	46	2	48
elastic_pathing	F-M-Hm	few-shot	50	1,959	2	62	4	32
elastic_pathing	F-M-Hm	few-shot	100	1,959	2	49	4	45

TABLE XVI

COMPLEMENTARITY OF THE MAP METHODS (FIVE-MINUTE WINDOWS): PER-WINDOW ORACLES OVER A METHOD SET (LOWEST ADE; ANY ROUTE RECOVERED), THE SHARE OF WINDOWS EVERY MEMBER FAILS, AND THE FIXED REFERENCE METHOD.

set	fold	n	oracle median m [CI]	oracle route % [CI]	all fail % [CI]	reference	ref. median	ref. route %
A steering map, heading known	test	1,917	25 [18, 35]	97 [95, 98]	3.1 [2.1, 5.2]	beam	37	88
A steering map, heading known	few-shot	1,959	15 [11, 28]	98 [97, 99]	1.9 [1.2, 2.8]	beam	21	92
B speed-only map, heading known	test	1,917	50 [25, 93]	78 [68, 86]	22.3 [14.3, 32.5]	beam_none	165	54
B speed-only map, heading known	few-shot	1,959	40 [21, 67]	77 [70, 82]	22.8 [18.2, 29.9]	beam_none	343	47
B' speed-only map, heading unknown	test	1,917	61 [29, 139]	72 [59, 81]	28.4 [19.0, 40.5]	beam_none_hm	1,314	32
B' speed-only map, heading unknown	few-shot	1,959	54 [36, 81]	70 [64, 74]	30.1 [25.7, 36.2]	beam_none_hm	1,527	29
C published, heading known	test	1,917	30 [21, 46]	94 [90, 95]	6.3 [4.9, 9.5]	nan	–	–
C published, heading known	few-shot	1,959	17 [13, 36]	95 [93, 97]	4.7 [3.4, 7.5]	nan	–	–
D all map, heading known	test	1,917	24 [18, 34]	98 [96, 99]	2.3 [1.4, 3.6]	beam	37	88
D all map, heading known	few-shot	1,959	14 [11, 26]	98 [97, 99]	1.6 [0.9, 2.5]	beam	21	92
P pedal subset, heading unknown	test	245	40 [29, 70]	95 [86, 97]	4.9 [2.7, 14.1]	beam_hm	46	88
P pedal subset, heading unknown	few-shot	162	19 [14, 35]	93 [87, 98]	7.4 [2.4, 13.3]	beam_hm	21	86

TABLE XVII

COMMON FAILURE OF THE STEERING AND SPEED MAP METHODS BY REFERENCE TURN COUNT AND ROAD DENSITY (FOLDS POOLED).

stratum	level	n	drivers	all fail % [CI]	oracle route %	beam route %
reference turns	0-1	2,234	81	0.1 [0.0, 0.2]	100	98
reference turns	2-3	712	72	1.4 [0.6, 2.9]	99	88
reference turns	4-6	528	74	5.3 [3.3, 8.7]	95	73
reference turns	7+	402	66	8.7 [5.3, 12.4]	91	67
density	low (56/246 per km2 cuts)	1,292	33	0.9 [0.2, 1.7]	99	97
density	mid (56/246 per km2 cuts)	1,292	82	1.9 [1.2, 2.8]	98	87
density	high (56/246 per km2 cuts)	1,292	81	2.9 [1.6, 4.5]	97	86
all	all	3,876	92	1.9 [1.4, 2.6]	98	90

TABLE XVIII

PER-DRIVER ROUTE RECOVERY SHARE (FIVE-MINUTE WINDOWS; DRIVERS WITH AT LEAST FIVE WINDOWS): POOLED RATE, QUANTILES OVER DRIVERS AND THE SHARE OF DRIVERS BELOW 50 %.

method	fold	windows	drivers	≥ 5	pooled %	q10	q25	q50	q75	q90	below 50 %
pf	test	1,917	42	36	87	62	77	87	96	100	3
pf	few-shot	1,959	50	41	90	73	78	92	96	100	0
beam	test	1,917	42	36	88	62	75	88	100	100	3
beam	few-shot	1,959	50	41	92	77	83	94	97	100	0
opd2	test	1,917	42	36	79	54	64	78	84	89	8
opd2	few-shot	1,959	50	41	78	54	67	79	88	91	5
macro	test	1,917	42	36	62	21	33	50	73	82	50
macro	few-shot	1,959	50	41	65	32	42	61	77	84	37
ep_hp	test	1,917	42	36	58	23	31	41	64	71	61
ep_hp	few-shot	1,959	50	41	59	30	36	50	67	71	46
dewri_hp	test	1,917	42	36	62	15	26	45	61	78	53
dewri_hp	few-shot	1,959	50	41	59	17	30	56	70	85	49
spy_hp	test	1,917	42	36	20	0	0	10	18	33	100
spy_hp	few-shot	1,959	50	41	22	0	4	14	27	35	100
pf_none	test	1,917	42	36	56	9	22	35	58	66	61
pf_none	few-shot	1,959	50	41	48	12	29	50	64	75	49
beam_none	test	1,917	42	36	54	8	20	31	57	67	64
beam_none	few-shot	1,959	50	41	47	15	30	50	62	71	49
ep	test	1,917	42	36	53	14	25	34	51	69	72
ep	few-shot	1,959	50	41	50	23	32	43	57	67	59
dewri	test	1,917	42	36	46	9	17	35	50	63	72
dewri	few-shot	1,959	50	41	43	17	21	38	47	63	78

TABLE XIX

ROAD-DENSITY STRATA AROUND THE START (TERTILES; FOLDS POOLED): MEDIAN ADE WITH CI, 100 M COVERAGE AND TAIL-LOST SHARE.

method	protocol	window	tertile	n	drivers	median ADE m [CI]	c100 %	tail lost %
map_pf_steer__G3	O-M-Hp	5 min	low (¡ 56/km2)	1,292	33	29 [18, 59]	90	17
map_pf_steer__G3	O-M-Hp	5 min	mid (56 to 246/km2)	1,292	82	39 [27, 48]	83	27
map_pf_steer__G3	O-M-Hp	5 min	high (¿ 246/km2)	1,292	81	36 [26, 45]	83	27
map_pf_steer__G3	O-M-Hp	15 min	low (¡ 56/km2)	414	27	89 [46, 185]	65	53
map_pf_steer__G3	O-M-Hp	15 min	mid (56 to 246/km2)	381	65	202 [132, 373]	50	70
map_pf_steer__G3	O-M-Hp	15 min	high (¿ 246/km2)	374	61	264 [169, 564]	51	67
map_beam_steer__G3	F-M-Hp	5 min	low (¡ 56/km2)	1,292	33	24 [16, 53]	91	17
map_beam_steer__G3	F-M-Hp	5 min	mid (56 to 246/km2)	1,292	82	32 [23, 42]	85	25
map_beam_steer__G3	F-M-Hp	5 min	high (¿ 246/km2)	1,292	81	33 [22, 40]	84	27
map_beam_steer__G3	F-M-Hp	15 min	low (¡ 56/km2)	414	27	77 [44, 167]	67	53
map_beam_steer__G3	F-M-Hp	15 min	mid (56 to 246/km2)	381	65	163 [105, 326]	53	70
map_beam_steer__G3	F-M-Hp	15 min	high (¿ 246/km2)	374	61	162 [100, 275]	57	61
map_pf_none__G1	O-M-Hp	5 min	low (¡ 56/km2)	1,292	33	73 [32, 537]	73	37
map_pf_none__G1	O-M-Hp	5 min	mid (56 to 246/km2)	1,292	82	677 [234, 953]	49	62
map_pf_none__G1	O-M-Hp	5 min	high (¿ 246/km2)	1,292	81	843 [627, 978]	40	70
map_pf_none__G1	O-M-Hp	15 min	low (¡ 56/km2)	414	27	1,023 [96, 5,289]	42	77
map_pf_none__G1	O-M-Hp	15 min	mid (56 to 246/km2)	381	65	3,457 [2,459, 4,268]	23	86
map_pf_none__G1	O-M-Hp	15 min	high (¿ 246/km2)	374	61	3,468 [2,920, 4,276]	19	86
dr_steer__G3	O-N-Hp	5 min	low (¡ 56/km2)	1,296	33	2,102 [1,067, 4,256]	15	99
dr_steer__G3	O-N-Hp	5 min	mid (56 to 246/km2)	1,292	82	1,245 [819, 1,859]	22	95
dr_steer__G3	O-N-Hp	5 min	high (¿ 246/km2)	1,292	81	632 [428, 1,132]	30	90
dr_steer__G3	O-N-Hp	15 min	low (¡ 56/km2)	418	27	8,506 [6,751, 12,469]	5	99
dr_steer__G3	O-N-Hp	15 min	mid (56 to 246/km2)	381	65	6,461 [4,875, 7,873]	9	99
dr_steer__G3	O-N-Hp	15 min	high (¿ 246/km2)	374	61	3,373 [1,772, 5,018]	13	97
dr_curvature__G5	O-N-Hp	5 min	low (¡ 56/km2)	1,296	33	85 [74, 109]	65	67
dr_curvature__G5	O-N-Hp	5 min	mid (56 to 246/km2)	1,292	82	86 [76, 102]	63	63
dr_curvature__G5	O-N-Hp	5 min	high (¿ 246/km2)	1,292	81	70 [64, 82]	69	55
dr_curvature__G5	O-N-Hp	15 min	low (¡ 56/km2)	418	27	516 [479, 724]	25	94
dr_curvature__G5	O-N-Hp	15 min	mid (56 to 246/km2)	381	65	548 [443, 675]	25	95
dr_curvature__G5	O-N-Hp	15 min	high (¿ 246/km2)	374	61	418 [350, 492]	31	89

TABLE XX

EXTERNAL CHECK ON THE CONTROLLED SUPPLEMENT (ONE VEHICLE, ONE METROPOLITAN AREA, OTHER DRIVERS; FIVE-MINUTE WINDOWS), METHODS UNCHANGED AFTER FREEZING.

method	protocol	n	drives	drivers	fail	median ADE m [CI]	p90 m	route % [CI]
steering DR, no map	O-N-Hp	46	27	12	0	117 [93, 170]	449	7 [0, 10]
map PF, steering	O-M-Hp	46	27	12	0	75 [46, 492]	896	59 [26, 69]
MacroTracking	O-M-Hp	46	27	12	0	19 [14, 24]	144	85 [74, 95]
map beam, steering	F-M-Hp	46	27	12	0	46 [34, 308]	846	61 [37, 67]
OPD-II	F-M-Hp	46	27	12	0	25 [11, 270]	600	65 [32, 75]
map PF, speed	O-M-Hp	46	27	12	0	762 [503, 1,011]	1,545	9 [0, 20]
map beam, speed	F-M-Hp	46	27	12	0	763 [321, 890]	1,653	4 [0, 14]
Elastic Pathing, heading	F-M-Hp	46	27	12	0	337 [239, 779]	1,185	28 [6, 38]
Dewri et al., heading	F-M-Hp	46	27	12	0	605 [345, 670]	1,621	15 [0, 27]

TABLE XXI

MAP-EXTENT CHECKS: SHARE OF WINDOWS WHOSE REFERENCE ROUTE LEAVES THE 5 KM AREA ADAPTATION SQUARE OR THE VISIBLE DISC.

check	method	window	fold	n	out	share %	median max m
out_square	ct	5 min	test	245	181	73.9	4,486
out_square	ct	5 min	few-shot	162	106	65.4	3,822
out_square	spy	5 min	test	1,917	1,531	79.9	5,728
out_square	spy	5 min	few-shot	1,959	1,544	78.8	4,702
out_disc	G3	5 min	test	1,917	0	0.0	6,311
out_disc	G3	5 min	few-shot	1,959	0	0.0	5,168
out_disc	G3	15 min	test	608	0	0.0	17,222
out_disc	G3	15 min	few-shot	561	0	0.0	14,538
out_disc	G3	whole drive	test	282	18	6.4	10,190
out_disc	G3	whole drive	few-shot	454	37	8.1	6,263
out_disc	G1	5 min	test	1,917	0	0.0	6,311
out_disc	G1	5 min	few-shot	1,959	0	0.0	5,168
out_disc	G1	15 min	test	608	0	0.0	17,222
out_disc	G1	15 min	few-shot	561	0	0.0	14,538
out_disc	G1	whole drive	test	282	18	6.4	10,190
out_disc	G1	whole drive	few-shot	454	37	8.1	6,263

TABLE XXII

AGENT STUDY (SUPPLEMENTARY COHORT, ONE FIVE-MINUTE WINDOW PER HELD-OUT DRIVE): EVERY CONFIGURATION AND SEED, WITH THE SELECTOR FALLBACK ROW AND THE POOL ORACLE.

row	fold	n	route % [CI]	median ADE m [CI]	no out. %	cost s	calls	tokens
map beam (A)	test	268	69 [58, 85]	49 [29, 66]	0.0	0.3	–	–
supervised selector (B)	test	268	75 [67, 87]	38 [29, 50]	0.0	1.1	3.0	–
Qwen agent, seed 0	test	268	75 [66, 88]	41 [29, 55]	0.0	35.5	2.9	1,703
Qwen agent, seed 1	test	268	75 [65, 87]	44 [27, 54]	0.0	37.8	2.9	1,753
Qwen agent, seed 2	test	268	75 [66, 88]	39 [29, 52]	0.0	35.9	3.0	1,702
Qwen agent (C), seeds pooled	test	268	75 [66, 88]	41 [29, 53]	0.0	36.4	2.9	1,719
Qwen agent with selector fallback	test	268	75 [66, 88]	41 [29, 53]	0.0	–	–	–
pool oracle (diagnostic)	test	268	83 [74, 93]	27 [19, 32]	0.0	–	–	–
map beam (A)	few-shot	391	83 [76, 90]	24 [18, 36]	0.0	0.3	–	–
supervised selector (B)	few-shot	391	83 [78, 90]	24 [18, 34]	0.0	1.0	3.0	–
Qwen agent, seed 0	few-shot	391	84 [77, 91]	23 [18, 34]	0.0	31.9	3.0	1,626
Qwen agent, seed 1	few-shot	391	84 [77, 90]	23 [18, 34]	0.0	37.9	2.9	1,741
Qwen agent, seed 2	few-shot	391	84 [78, 91]	22 [18, 35]	0.0	35.2	2.9	1,711
Qwen agent (C), seeds pooled	few-shot	391	84 [77, 91]	23 [18, 34]	0.0	35.0	2.9	1,693
Qwen agent with selector fallback	few-shot	391	84 [77, 91]	23 [18, 34]	0.0	–	–	–
pool oracle (diagnostic)	few-shot	391	92 [87, 96]	16 [11, 23]	0.0	–	–	–

TABLE XXIII

AGENT STUDY, PAIRED COMPARISONS ON THE SAME WINDOWS: ROUTE RECOVERY DIFFERENCE IN PERCENTAGE POINTS (SEEDS POOLED INSIDE THE DRIVER RESAMPLES; PER-SEED VALUES) AND THE MEDIAN ADE DIFFERENCE ON WINDOWS WITH OUTPUT ON BOTH SIDES.

pair	fold	n	Δ route pp [CI]	per seed	Δ median ADE m [CI]
agent_vs_selector	test	268	+0.1 [-1.6, +3.4]	+0.4, -0.4, +0.4	0.0 [0.0, 0.0]
agent_vs_beam	test	268	+5.7 [+1.6, +8.5]	+6.0, +5.2, +6.0	0.0 [0.0, 0.0]
agent_vs_selector	few-shot	391	+0.7 [-2.6, +2.8]	+0.8, +0.5, +0.8	0.0 [0.0, 0.0]
agent_vs_beam	few-shot	391	+1.2 [-1.3, +3.8]	+1.3, +1.0, +1.3	0.0 [0.0, 0.0]

TABLE XXIV
SUPERVISED SELECTOR ON THE VALIDATION COHORT (CROSS-FITTED BY DRIVER).

row	windows	drivers	route % [CI]
selector_cv	610	43	82 [77, 87]
beam1	610	43	79 [72, 84]
pf1	610	43	75 [69, 80]
ep1	610	43	28 [21, 40]
oracle	610	43	91 [88, 94]

TABLE XXV
VALIDATION PILOT OF THE AGENT (40 WINDOWS, THREE SEEDS).

tag	n	valid %	trunc. %	forced %	turns	calls	tokens	thinking	LLM s
agent_qwen3_s0	40	100.0	2.5	87.5	3.8	3.0	1,703	1,521	27.9
agent_qwen3_s1	40	100.0	0.0	85.0	3.8	2.9	1,754	1,577	38.0
agent_qwen3_s2	40	100.0	0.0	77.5	3.7	2.9	1,693	1,513	35.6
all	120	100.0	0.8	83.3	3.8	2.9	1,717	1,537	34.6